

Rating manual

Observer-rated user behavior in human-robot interaction dialogs

1. Purpose of the rating task

This rating task assesses the **human user's observable conversational behavior** in short human-robot interaction dialogs. Raters should evaluate how the user participates in the conversation based only on the transcript. The goal is not to judge the robot, the topic, or the rater's personal impression of the robot, but to assess what the human user visibly says, asks, reveals, and expresses during the dialog.

2. Unit of analysis

Each complete dialog is one rating unit. Read the whole dialog before rating. Some dimensions depend on the overall pattern of the user's behavior rather than on one isolated turn.

3. General rating rules

- Rate only the **human user's behavior**. Do not directly evaluate the robot's answer quality, intelligence, appearance, or personality.
- Base your ratings only on what is visible in the dialogs. Do not infer hidden emotions, intentions, or personality traits beyond what the user expresses.
- Transcription errors should be dealt with as leniently as possible; that is, if a statement can still be understood, the transcription errors should not be included in the evaluation.
- For short dialogs, rate the quality of the available user contribution. Do not automatically give a low score only because the dialog is short, unless the shortness reflects minimal, passive, or disengaged user behavior.
- The anchor examples are illustrative. Raters should score the whole dialog, not only isolated phrases.

4. Per-dimension rating scales

Dimensions at a glance:

1. User engagement
2. User self-disclosure
3. User topical alignment
4. User elaboration
5. User initiative
6. User politeness
7. User frustration / dissatisfaction

8. Overall user-side interaction quality

Dimension 1: User engagement		
Question:		To what extent does the user appear engaged, interested, and motivated to keep the conversation going?
Definition:		User engagement refers to the user’s visible involvement in the interaction, shown through willingness to answer, positive or curious tone, continued participation, and motivation to keep the conversation going. Do not infer engagement from dialog length alone.
Rating scale & anchor examples: Assume the robot asks: “Would you like to tell me what you usually eat for breakfast?”		
1	Clearly disengaged	“I don’t want to talk about breakfast.”
2	Low engagement	“Breakfast is okay.”
3	Neutral / basic engagement	“I usually eat breakfast in the morning.”
4	Engaged	“I usually eat breakfast in the morning, and I like having some time for it.”
5	Highly engaged	“I actually enjoy talking about breakfast routines. Mine changes a bit during the week, and I like thinking about what makes a good start to the day.”

Dimension 2: User self-disclosure	
Question:	To what extent does the user reveal personal information, preferences, experiences, feelings, routines, or reasons?
Definition:	User self-disclosure refers to how much personal content the user shares. Minimal preferences count as low self-disclosure; specific routines, experiences, reasons, or personally meaningful details indicate higher self-disclosure.
Rating scale & anchor examples: Assume the robot asks: “What do you usually eat for breakfast, and what is your usual breakfast routine?”	

1	No personal disclosure	"People eat different things for breakfast."
2	Minimal personal fact / preference	"I eat bread for breakfast."
3	Basic personal routine / experience	"I usually eat bread with cheese in the morning."
4	Specific personal context	"I usually eat bread with cheese before work because mornings are rushed for me."
5	Rich personal disclosure	"Breakfast is usually stressful for me during the week because I am in a hurry, but on weekends I like eating slowly with my family."

Dimension 3: User topical alignment		
Question:		To what extent are the user's contributions relevant to the current topic, question, or conversational task?
Definition:		User topical alignment refers to how well the user stays connected to the selected topic and the robot's current prompt. Reasonable clarification or topic-related redirection can still be aligned; unrelated, disruptive, or incomprehensible turns should be rated lower.
Rating scale & anchor examples: Assume the robot asks: "What do you usually eat for breakfast?"		
1	Unrelated / incomprehensible	"Who is the best AI?"
2	Weakly aligned	"Breakfast is fine, but can we talk about AI instead?"
3	Mixed / basic alignment	"Maybe bread or something. Anyway, I also like watches."
4	Mostly aligned	"I usually eat bread with cheese for breakfast."

5	Fully aligned and responsive	"On weekdays I usually eat bread with cheese and coffee, and on weekends I sometimes have something sweet like Nutella."
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Dimension 4: User elaboration		
Question:		To what extent does the user provide informative, specific, and sufficiently elaborated answers?
Definition:		User elaboration refers to the amount and usefulness of information provided by the user, including details, explanations, examples, reasons, or concrete descriptions. This rewards specificity and useful detail , but it does not require emotional or personally meaningful disclosure.
Rating scale & anchor examples: Assume the robot asks: "What do you usually eat for breakfast?"		
1	Non-informative	"Yes."
2	Minimal detail	"Bread."
3	Basic information	"I eat bread with cheese for breakfast."
4	Specific and informative	"I eat bread with cheese and drink coffee, usually before work."
5	Rich, detailed, multi-faceted	"On weekdays I usually have bread with cheese and coffee; on weekends I sometimes add eggs or something sweet, depending on how much time I have."

Dimension 5: User initiative		
Question:		To what extent does the user actively shape, advance, or repair the conversation beyond simply answering questions?
Definition:		User initiative refers to the user's active role in the interaction, such as asking follow-up questions,

		requesting clarification, proposing a direction, correcting misunderstandings, or making the conversation more interactive.
Rating scale & anchor examples: Assume the robot asks: “What do you usually eat for breakfast?”		
1	Passive/no initiative	“Okay.”
2	Mostly reactive	“I eat bread.”
3	Occasional mild initiative	“I eat bread. Maybe we can also talk about weekend breakfast.”
4	Actively contributes	“I eat bread during the week, but weekends are different. Should I describe both?”
5	Strongly proactive	“I can compare my weekday and weekend breakfasts, and I’d also like to ask what kinds of breakfasts other people usually mention.”

Dimension 6: User politeness		
Question:		To what extent is the user’s wording respectful, socially appropriate, and considerate?
Definition:		User politeness refers to the user’s observable interpersonal tone, including polite markers, respectful phrasing, patience, and absence of insults or hostility. Neutral direct answers should receive a neutral score, not automatically a high score.
Rating scale & anchor examples: Assume the robot asks politely: “Could you tell me what you usually eat for breakfast?”		
1	Rude / hostile	“This is stupid. Stop asking me about breakfast.”
2	Abrupt / mildly rude	“Just ask the next question.”
3	Neutral / task-focused	“I eat bread with cheese.”

4	Polite / respectful	"I usually eat bread with cheese, thank you for asking."
5	Very polite / appreciative	"Yes, of course. I usually eat bread with cheese, and thank you, this is a nice question."

Dimension 7: User frustration / dissatisfaction		
Question:	To what extent does the user visibly express frustration, dissatisfaction, annoyance, impatience, or unwillingness to continue the interaction?	
Definition:	User frustration / dissatisfaction refers to observable negative reactions toward the conversation, the robot's behavior, repeated questions, misunderstandings, or the interaction process. This includes complaints, irritated wording, impatience, negative evaluations, or refusal to continue. Simple clarification or uncertainty should not be rated as frustration unless it is expressed with visible annoyance, dissatisfaction, or complaint.	
Rating scale & anchor examples: Assume the robot repeats: "What do you usually eat for breakfast?" after the user has already responded.		
1	No visible frustration	"I usually eat bread with cheese for breakfast."
2	Slight confusion / dissatisfaction	"This question is a little unclear, but I usually eat bread with cheese."
3	Mixed / moderate frustration	"I already answered this, but okay. I usually eat bread with cheese."
4	Clear frustration	"This is frustrating because I already answered the breakfast question."
5	Strong frustration / refusal	"This is really annoying. Stop asking me about breakfast. I don't want to continue."

Dimension 8: Overall user-side interaction quality		
Question:		Overall, how successful is the user's conversational participation in this dialog?
Definition:		Overall user-side interaction quality is a holistic judgment of the user's observable contribution, integrating engagement, relevance, informativeness, initiative, social tone, intelligibility, and lack of disruptive frustration. Do not directly rate the robot; only consider how the user participates.
Rating scale & anchor examples:		
1	Poor	The user's participation is mostly unusable, unrelated, hostile, incomprehensible, or refuses the interaction.
2	Low	The user contributes very little; responses are mostly minimal, unclear, weakly relevant, or socially abrupt.
3	Adequate / mixed	The dialog functions, but user participation is basic, uneven, or only partly informative.
4	Good	The user contributes clearly, relevantly, and appropriately across most of the dialog, with minor issues only.
5	Excellent	The user's participation is consistently clear, relevant, informative, engaged, and helps the dialog flow smoothly.